Technische Universität Dresden

Data Visualization
Course Work

1 – Dear Data



Computergraphik und Visualisierung



INTERACTIVE MEDIA LAB DRESDEN

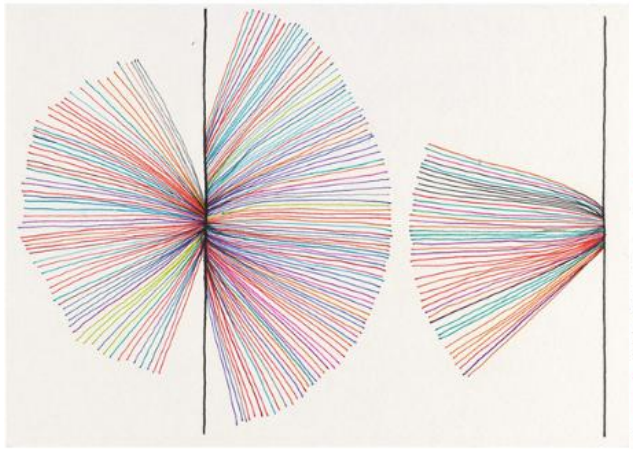
Faculty of Computer Science – Institute of Software and Multimedia Technology

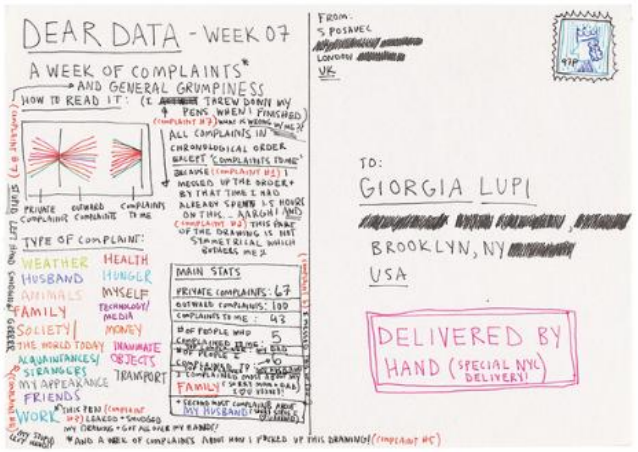
Course Work – Assignment 1
1 Dear Data

Collect and visualize your own data:

- http://www.dear-data.com/theproject
- Take a week to collect data:
  Count, measure, observe…
- Draw a visualization using pen & paper
- (At least) one visualization per person!





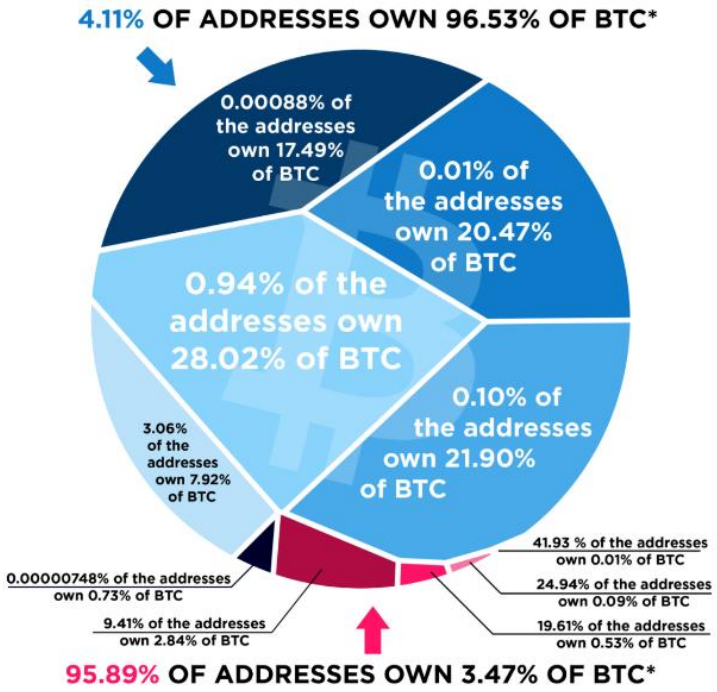


- Carefully think about your design decisions
- Which type of visualization?
- How to encode data to visual variables?
- Use of color?

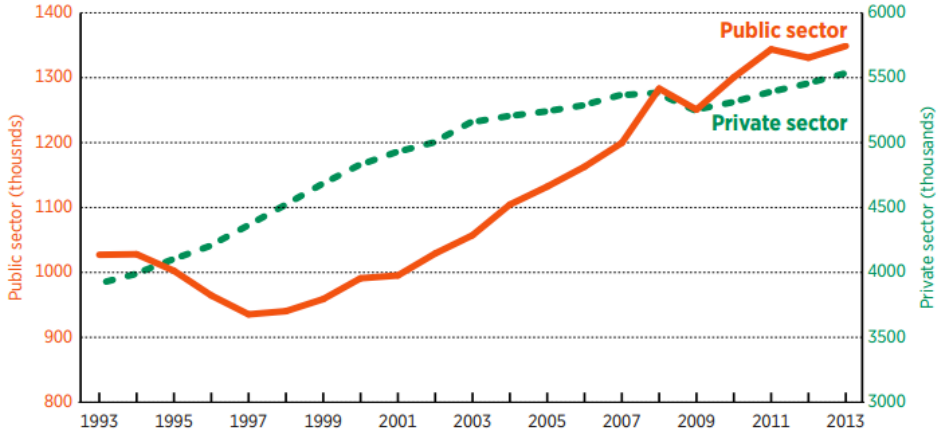
Course Work – Assignment 1
2.1 Lecture Recap: VisLies

Visualizations can be misleading!

- Discuss the quality of visualizations…
- What's the main problem for each of the visualizations?
- How could you change the visualization to eliminate these problems?



Source: https://howmuch.net/articles/bitcoin-wealth-distribution



Source: https://www.fraserinstitute.org/sites/default/files/ontario-no-longer-a-place-toprosper.pdf

Course Work – Assignment 1
2.1 Lecture Recap: VisLies – an example

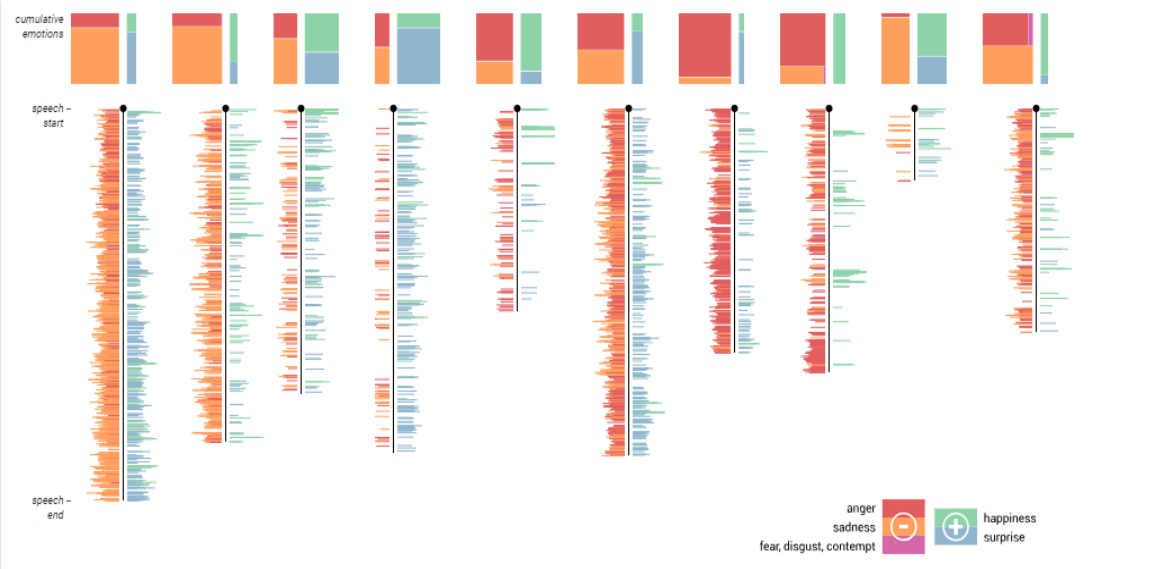


Source: https://www.bizbuysell.com/blog/buying-or-selling-a-small-business-is-a-big-deal/

Course Work – Assignment 1
2.2 Lecture Recap: Perception and Color

Gestalt Principles:

- Which gestalt principles have been used in the visualization shown below?
- Name them, and explain briefly!



Color Scales:

- Rainbow color scale has several problems
- Name & explain at least three of them
- Give examples of better scales
- visit https://sciviscolor.org/



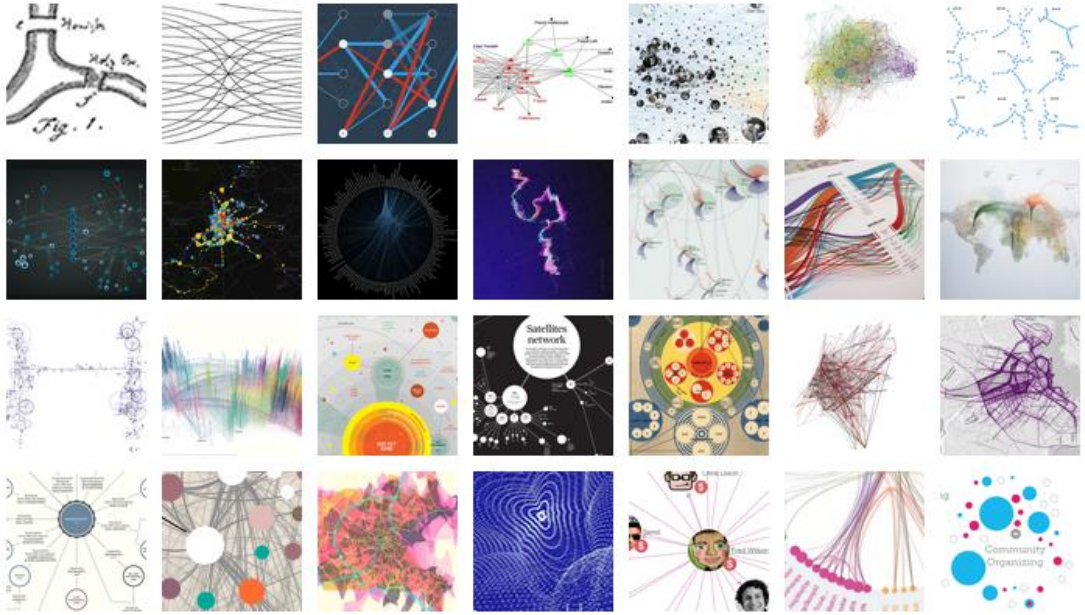
Course Work – Assignment 1 Additional Reads

Color Schemes

- Good source for ready-to-use color schemes: http://colorbrewer2.org/
- Creating own schemes with Cologorical (check the paper for an in-depth read about it) http://vrl.cs.brown.edu/color
- Post on multi-hue color scales: https://www.vis4.net/blog/2013/09/mastering-multi-hued-color-scales/

Blogs on Visualization Design

- "The little of visualization design" by Andy Kirk: http://www.visualisingdata.com/2016/03/little-visualisation-design/
- Eagereyes (Robert Kosara): https://eagereyes.org/
- Data sketches: http://www.datasketch.es/
- Flowingdata: https://flowingdata.com/



Visualization Showcases

- Visual Complexity: http://www.visualcomplexity.com/vc/
- And… big newspapers!
  - NY Times: https://twitter.com/nytgraphics
  - Washington Post: https://twitter.com/postgraphics

Course Work – Assignment 1 Submission

Submission of all files to OPAL

- TXT file with your answers to the questions.
- Scan or photo of your Dear Data sketch as JPG, PNG, or PDF
- 1-min video explaining your Dear Data sketch as MP4 (1080p, H.264, <50MB)